Integral Challenge Problems

1. $\displaystyle\int \left(\sin^{-1} x\right)^2 \, dx$
2. $\displaystyle\int x \sin^{-1} x \, dx$
3. $\displaystyle\int \sin^{-1} \sqrt{x} \, dx$
4. $\displaystyle\int \frac{1}{1 - \tan^2 x} \, dx$
5. $\displaystyle\int \ln\left(\sqrt{x} + \sqrt{x+1}\right) \, dx$
6. $\displaystyle\int \frac{1}{x - \sqrt{1 - x^2}} \, dx$
7. $\displaystyle\int \frac{2e^{2x} - e^x}{\sqrt{3e^{2x} - 6e^x - 1}} \, dx$
8. $\displaystyle\int \frac{1}{x^6 - 1} \, dx$
9. $\displaystyle\int \frac{1}{x^4 + 4} \, dx$
10. $\displaystyle\int \frac{1}{x(x+1)(x+2)\cdots(x+n)} \, dx$